

CS & IT ENGINEERING



Computer Network

IPv4 Header

Lecture No. - 06

By - Abhishek Sir





Recap of Previous Lecture



Topic

Fragmentation & Reassembly





Topics to be Covered



Topic

Time-to-Live

Topic

Protocol Type

Topic

Checksum

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW



#Q. Which of the following statements about IPv4 fragmentation is/are TRUE?

- ☒ A The fragmentation of an IP datagram is performed only at the source of the datagram FALSE
- ☒ B The fragmentation of an IP datagram is performed at any IP router which finds that the size of the datagram to be transmitted exceeds the MTU TRUE
- ☒ C The reassembly of fragments is performed only at the destination of the datagram TRUE
- ☒ D The reassembly of fragments is performed at all intermediate routers along the path from the source to the destination FALSE

Ans: B & C

[MSQ]

[GATE-2021][2 Mark]



#Q. Consider two hosts P and Q connected through a router R. The maximum transfer unit (MTU) value of the link between P and R is 1500 bytes, and between R and Q is 820 bytes.

A TCP segment of size 1400 bytes was transferred from P to Q through R, with IP identification value as 0x1234. Assume that the IP header size is 20 bytes. Further, the packet is allowed to be fragmented, i.e., Don't Fragment (DF) flag in the IP header is not set by P.

Which of the following statements is/are correct?

- ☒ A Two fragments are created at R and the IP datagram size carrying the second fragment is 620 bytes. TRUE
- ☐ B If the second fragment is lost, R will resend the fragment with the IP identification value 0x1234. FALSE
- ☒ C If the second fragment is lost, P is required to resend the whole TCP segment. TRUE
- ☐ D TCP destination port can be determined by analysing only the second fragment. FALSE

Ans: A & C

$$\text{TCP Segment Size} = 1400 \text{ bytes}$$

$$\text{IPv4 Header Size} = 20 \text{ bytes}$$

$$\text{MTU}_1 = 1500 \text{ bytes}$$

$$\begin{aligned} \text{New Payload Size}_1 &= [\text{MTU} - \text{Header Size}] \text{ bytes} \\ &= (1500 - 20) \text{ bytes} \\ &= 1480 \text{ bytes} \end{aligned}$$

$$\text{MTU}_2 = 820 \text{ bytes}$$

$$\begin{aligned} \text{New Payload Size}_2 &= [\text{MTU} - \text{Header Size}] \text{ bytes} \\ &= (820 - 20) \text{ bytes} \\ &= 800 \text{ bytes} \end{aligned}$$



(20+) 1400 bytes
(TCP segment)

(20+) 800 bytes
Payload₁

(off = 0)
M = 1

(20+) 600 bytes
Payload₂

(M = 0)

TCP/UDP Header $\Rightarrow$ Source & Dest. Port No.



IPv4/IPv6 Header $\Rightarrow$ Source & Dest. IP Add.

Link Layer Frame Header $\Rightarrow$ Source & Dest. MAC Add.



Topic : IPv4 Packet Header



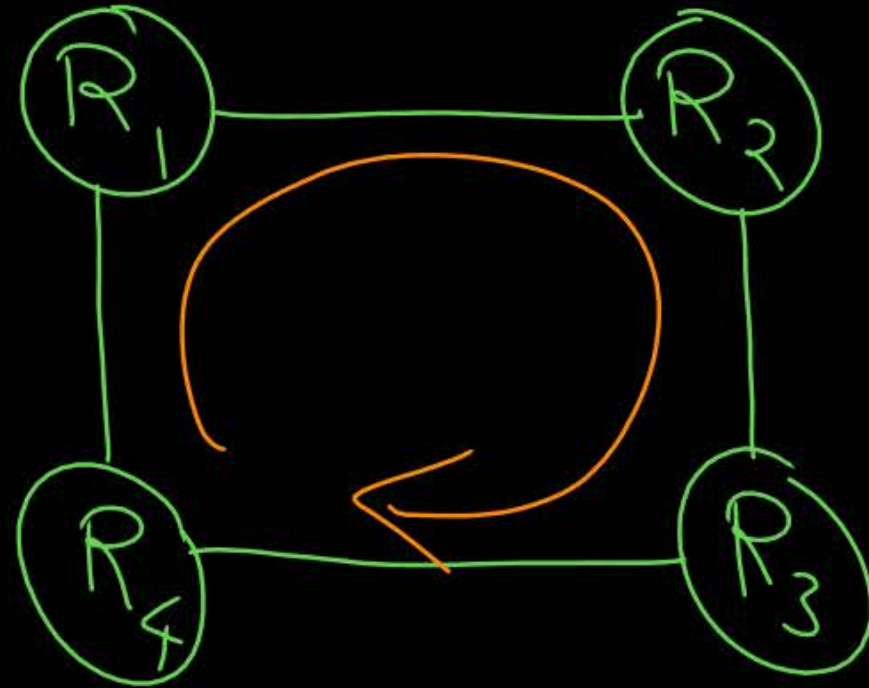
0			16		31
VER	HLEN	Type of Services	Total Length		
Identification Number			0	D F	M F
Time-to-Live		Protocol Type	Header Checksum		
Source IPv4 Address (32 - bits)					
Destination IPv4 Address (32 - bits)					
Optional Header (Options)					
Payload					



Topic : Time-to-Live



- Time-to-live (TTL) is 8-bit field $\Rightarrow$ [Range 0 to 255]
- Life time of an IP datagram
- Avoid indefinite traversing of an IP datagram in the network
[Router's may form circuit]



[MCQ]

IIT-KGP

[GATE-CS-2006] [1 Mark]



#Q. For which one of the following reasons does Internet Protocol (IP) use the time-to-live (TTL) field in the IP datagram header ?

- ☒ A Ensure packets reach destination within that time
- ☒ B Discard packets that reach later than that time
- ☒ C Prevent packets from looping indefinitely
- ☒ D Limit the time for which a packet gets queued in intermediate routers.

Ans: C

[MCQ]

11T-6

[GATE-2010] [1 Mark]



#Q. One of the header fields in an IP datagram is the Time to Live (TTL) field. Which of the following statements best explains the need for this field?

- ☒ A It can be used to prioritize packets
- ☒ B It can be used to reduce delays
- ☒ C It can be used to optimize throughput
- ☒ D It can be used to prevent packet looping

Ans: D



Topic : Time-to-Live



- Each intermediate router decrement TTL value by one // TTL--;
↑
- if TTL value decremented to zero // [TTL ≤ 0]
then router discard the datagram
and send ICMP error message "Time Exceeded" to source host
↑↑

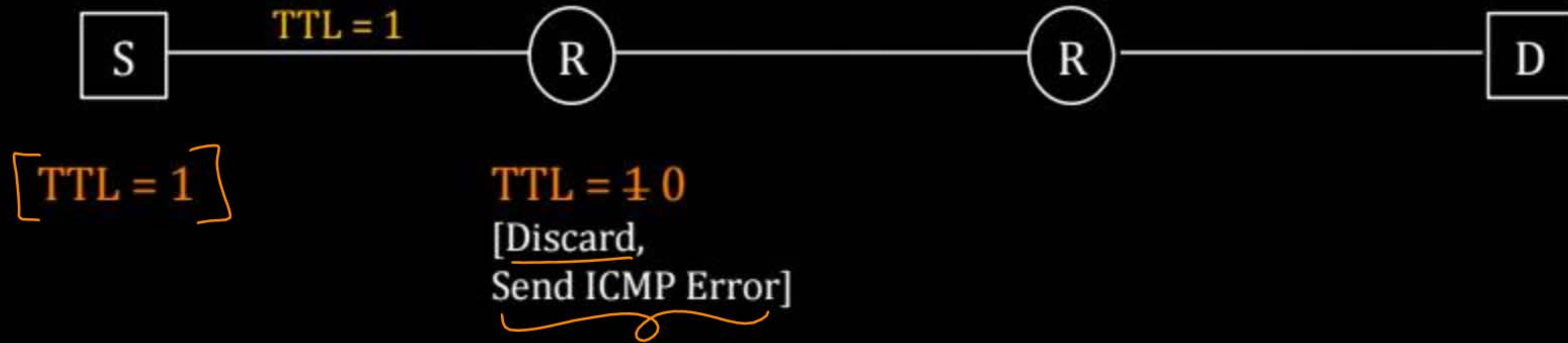


Topic : Time-to-Live



- Destination Host may decrement TTL value by one // TTL--;
- if TTL value decremented to less than zero // [TTL < 0]
then destination host discard the datagram
and send ICMP error message "Time Exceeded" to source host

CASE I :



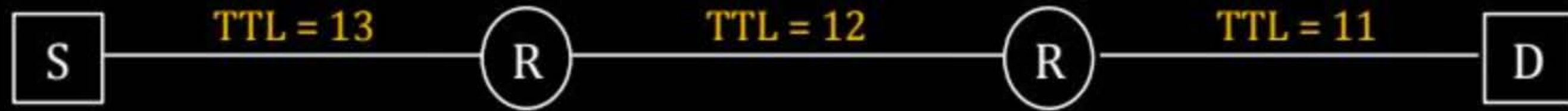
CASE II :



CASE III :



CASE IV :



[TTL = 13]

TTL = ~~13~~ 12

TTL = ~~12~~ 11

TTL = ~~11~~ 10

[Forward
to higher layer]

Accepted,

CASE V :



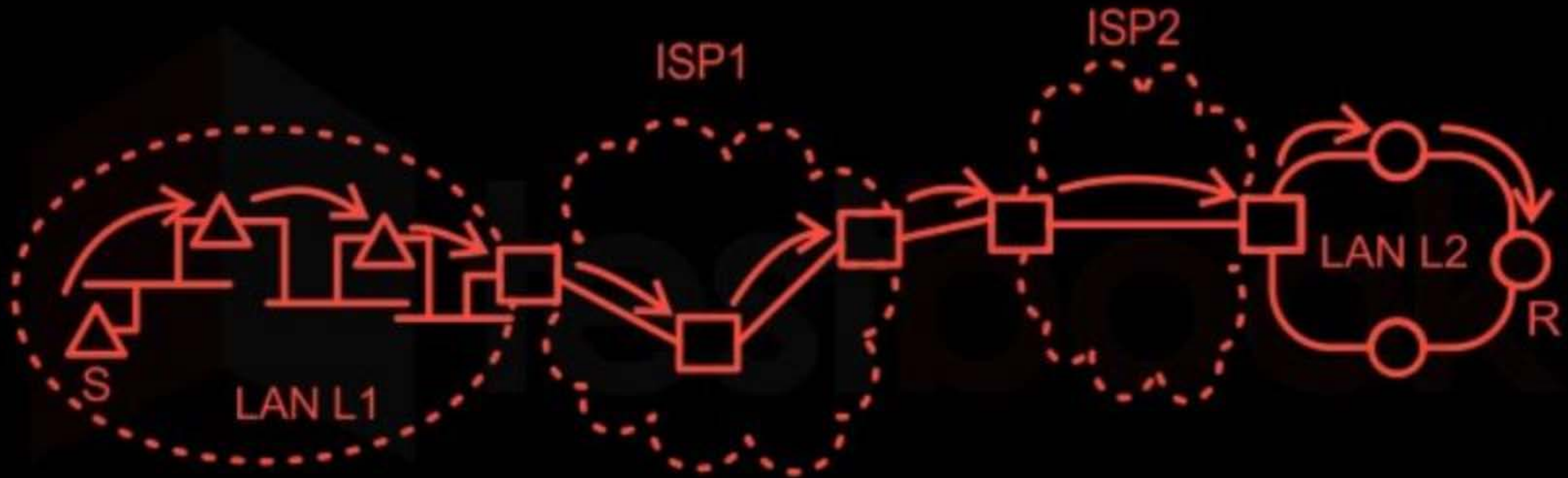
Note : TTL = 1, can use when source and destination host are directly connected.

[NAT] (H.W.)

[GATE-2014][1 Mark]



#Q. In the diagram shown below L1 is an Ethernet LAN and L2 is a Token-Ring LAN. An IP packet originates from sender S and traverses to R, as shown. The link within each ISP, and across two ISPs, are all point to point optical links. The initial value of TTL is 32. The maximum possible value of TTL field when R receives the datagram is ____.





Topic : Time-to-Live



→ Hop Limit 8-bit field in IPv6 Header



→ Same as Time-to-live (TTL) field of IPv4 Header



Topic : Protocol Type



- Protocol Type is 8-bit field in IPv4 Header
- Indicates higher-level protocol ←
- Indicates the type of protocol encapsulated in the payload
- Demultiplexing of protocol at receiver

(TCP segment)



(IPv4 Packet Payload)

TCP



IP



TCP



IP



(UDP segment)



IPv4 Packet Payload

UDP



IP



UDP



IP

(ICMP Error Packet)



(IPv4 Packet Payload)

ICMP



IP



ICMP



IP



Topic : Protocol Type



Number	Protocol
<u>1</u>	<u>ICMP</u>
<u>2</u>	<u>IGMP</u>
<u>6</u>	<u>TCP</u>
<u>17</u>	<u>UDP</u>
<u>89</u>	<u>OSPF</u>

[MCQ]

[GATE-2025] [1 Mark]



#Q. A machine receives an IPv4 datagram. The protocol field of the IPv4 header has the protocol number of a protocol X. Which ONE of the following is NOT a possible candidate for X?

☒ A Internet Control Message Protocol (ICMP) $ICMP \rightarrow IP$

☒ B Internet Group Management Protocol (IGMP) $IGMP \rightarrow IP$

☒ C Open Shortest Path First (OSPF) $OSPF \rightarrow IP$

☒ D Routing Information Protocol (RIP) $RIP \rightarrow UDP \rightarrow IP$

Ans. D

$DHCP \rightarrow UDP \rightarrow IP$
 $BGP \rightarrow TCP \rightarrow IP$



Topic : Header Checksum



→ Header Checksum is 16-bit field in IPv4 Header

→ [Internet Checksum]

→ Ensures the integrity of the IPv4 header during routing

↓
(including options, excluding payload)



Topic : Header Checksum



→ Calculated over the IPv4 header only
[including optional header]

→ Block Size is 16-bits

→ 16 bit one's complement of the one's complement sum of all 16 bit words
in the IPv4 header.



Topic : Header Checksum

Block Size = 16 bits

While computing the checksum, the value of the checksum field should be initialized with zero.



IPv4 Header

VER	HLEN	Type of Service
Total Length		
Identification Number		
Flag	Fragmentation Offset	
Time-to-Live		Protocol Type
Checksum [0 0 0 0 0 0 0 0]		
Source IP Address (16-bits)		
Source IP Address (16-bits)		
Destination IP Address (16-bits)		
Destination IP Address (16-bits)		



Topic : Checksum



- Error detection technique (No provision for error correction)
- Both sender and receiver must agree on same block size (n) (mostly $n=16$)
- n bit one's complement of the one's complement sum of all n bit words



Topic : Checksum



→ Sender generate (n-bit Checksum) from data blocks
and then send data along with checksum

→ Receiver check the “received data along with checksum” is balanced or not

→ While computing the checksum,
the value of the checksum field should be initialized with zero

→ While transmission,
checksum field should be updated with computed checksum



Block Size = n

Receiver

Block 1

Block 2

Block 6 Checksum

Block 10

^{1's comp.}
n - bit ~~binary~~ sum

1's complement

Result

Sender

Block 1

Block 2

Block 6 000 000

Block 10

^{1's comp.}
n - bit ~~binary~~ sum

1's complement

Checksum



Topic : Checksum



Result : [Computed at receiver]

if Result == ZERO :

then Receiver concluded “No any error detected”

else

Receiver concluded “Error detected”

Recv.

$$\begin{array}{r}
 12 \\
 07 \\
 15 \\
 27 \\
 \hline
 \boxed{38} \\
 \hline
 99
 \end{array}$$

↓ complement

$$(99 - 99) = 0$$

Sender

$$\begin{array}{r}
 12 \\
 07 \\
 15 \\
 27 \\
 \hline
 \boxed{100} \\
 \hline
 61
 \end{array}$$

↓ complement

$$(99 - 61) = 38$$

Example 1:-

Suppose block size = 5

3 → 11
5 → 101
4 → 100

10 11011
+
11011

11011
+ 11011
10
11101

Sender

10101
10110

01101

11011

00000

10011

01010

1011011

11101

checksum = 00010

Example 1:-

Suppose block size = 5

Receiver

10 10 11 1

1 0 1 1 0

0 1 1 0 1

1 1 0 1 1

0 0 0 1 0

1 0 0 1 1

0 1 0 1 0

10 11101

11111

↓ is comp.
00000

10 11101
+
11101
10
11111

Sender

1 0 1 1 0

0 1 1 0 1

1 1 0 1 1

0 0 0 0 0

1 0 0 1 1

0 1 0 1 0

10 11011

11101

is comp. sum ←
checksum ← 00010
is comp.

Example 1:-

Suppose block size = 5

Receiver

1 0 1 1 0

0 1 1 0 1

1 1 0 1 1

0 0 0 1 0

1 0 0 1 1

0 1 0 1 0

1 0 1 1 1 0 1

1 1 1 1 1

↓ 1's comp.

0 0 0 0 0

Sender

1 0 1 1 0

0 1 1 0 1

1 1 0 1 1

0 0 0 0 0

1 0 0 1 1

0 1 0 1 0

1 0 1 1 0 1 1

1 1 1 0 1

↓ 1's comp.

0 0 0 1 0

Example 2:-

Suppose block size = 5

$$\begin{array}{r}
 \begin{array}{ccccc}
 1 & 1 & 0 & 1 & 1 & 0 \\
 \hline
 & & 1 & 1 & 1 & 0 \\
 + & & 0 & 1 & 1 & 1 & 0 \\
 \hline
 1 & 0 & 0 & 0 & 1 &
 \end{array}
 \end{array}$$

Sender

11110

01111

11011

00000

11011

01011

110110

10001

↓ is comp.
01110

Example 2:-

Suppose block size = 5

Receiver

④ ① ④ ① ⑤

1 1 1 1 0

0 1 1 1 1

1 1 0 1 1

0 1 1 1 0

1 1 0 1 1

0 1 0 1 1

1 1 1 1 0 0

1 1 1 1 1

↓ 1's comp.
0 0 0 0 0

Sender

1 1 1 1 0

0 1 1 1 1

1 1 0 1 1

0 0 0 0 0

1 1 0 1 1

0 1 0 1 1

1 1 0 1 1 1 0

1's comp. sum ← 1 0 0 0 1

checksum ← 0 1 1 1 0

$$\begin{array}{r}
 8 \rightarrow 1000 \\
 9 \rightarrow 1001 \\
 \hline
 \begin{array}{ccccccc}
 1 & 1 & & 1 & 1 & 1 & 0 & 0 \\
 \hline
 & & & & & & \oplus & \\
 \hline
 & & & & & & &
 \end{array} \\
 \hline
 \begin{array}{r}
 + 11100 \\
 11 \\
 \hline
 111111
 \end{array}
 \end{array}$$

Example 2:-

Suppose block size = 5

Receiver

1 1 1 1 0

0 1 1 1 1

1 1 0 1 1

0 1 1 1 0

1 1 0 1 1

0 1 0 1 1

1 1 1 1 1 0 0

1 1 1 1 1



0 0 0 0 0

Sender

1 1 1 1 0

0 1 1 1 1

1 1 0 1 1

0 0 0 0 0

1 1 0 1 1

0 1 0 1 1

1 1 0 1 1 1 0

1 0 0 0 1



0 1 1 1 0



Topic : Checksum



Example 3 :-

String 1 = 1 0 0 1 0 1 0 0

String 2 = 1 1 0 0 1 1 0 1

Calculate checksum ?

Ans = 10011101



$$\begin{array}{r} 01100001 \\ + 01100001 \\ \hline 01100001 \end{array}$$

↓ is comp.
10011101 checksum



Topic : Checksum



- Checksum detect “**all single bit error**”
- In case of burst error, checksum able to detect “**all odd number of errors**”
- Checksum can be “**all zero bits**”,
but checksum can never be “**all one bits**”

Example 4:-

Suppose block size = 5

Sender

1 0 0 0 0

0 1 0 0 0

0 0 1 0 0

0 0 0 0 0

0 0 0 1 0

0 0 0 0 1

1 1 1 1 1

checksum = 0 0 0 0 0 ↓ 1's comp.

Example 5:-

Suppose block size = 5

Sender

0 0 0 0 0

0 0 0 0 0

0 0 0 0 0

0 0 0 0 0

0 0 0 0 0

0 0 0 0 0

0 0 0 0 0

↓ 1's comp



[MSQ]

H.W.

ISC

[GATE-2024][1 Mark]



#Q. Which of the following fields of an IP header is/are always modified by any router before it forwards the IP packet?

- ☐ A Source IP Address
- ☐ B Protocol
- ☐ C Time to Live (TTL)
- ☐ D Header Checksum

[MSQ] H.W.

11SC

[GATE-2024][1 Mark]



#Q. Which of the following fields is/are modified in the IP header of a packet going out of a network address translation (NAT) device from an internal network to an external network ?

- ☐ A Header Checksum
- ☐ B Source IP
- ☐ C Destination IP
- ☐ D Total Length

[MCQ] H.W.

IIT-K [GATE-2015][1 Mark]



#Q. Which one of the following fields of an IP header is NOT modified by a typical IP router ?

- ☐ A Checksum
- ☐ B Source address
- ☐ C Time to Live (TTL)
- ☐ D Length

[MCQ]

H.W.

IIT-KGP

[GATE-2014][1 Mark]



#Q. Host A (on TCP/IPv4 network A) sends an IP datagram D to host B (also on TCP/IPv4 network B). Assume that no error occurred during the transmission of D. When D reaches B, which of the following IP header field(s) may be different from that of the original datagram D ?

(i) TTL

(ii) Checksum

(iii) Fragment

A (i) only

B (i) and (ii) only

C (ii) and (iii) only

D (i), (ii) and (iii)

[MCQ]

11T-D

[GATE-IT-2004] [2 Mark]



#Q. In the TCP/IP protocol suite, which one of the following is NOT part of the IP header?

☒ A Fragment Offset

☒ B Source IP address

☒ C Destination IP address

☒ D Destination port number $\Rightarrow$ TCP/UDP Header

$\Rightarrow$ IP Header

Ans: D



2 mins Summary



Topic

Time-to-Live ✓

Topic

Protocol Type ✓

Topic

Checksum ✓



THANK - YOU

